RISOLVERE I SEGUENTI CIRCUITI, CALCOLANDO LE CORRENTI SUI VARI RAMI E LE TENSIONI AI NODI DEI SEGUENTI CIRCUITI UTILIZZANDO

a) TEORIA DELLE MAGLIE E DEI NODI
b) SOVRAPPOSIZIONE DEGLI EFFETTI
c) TEOREMA DI THEVENIN
d) TEOREMA DI NORTON

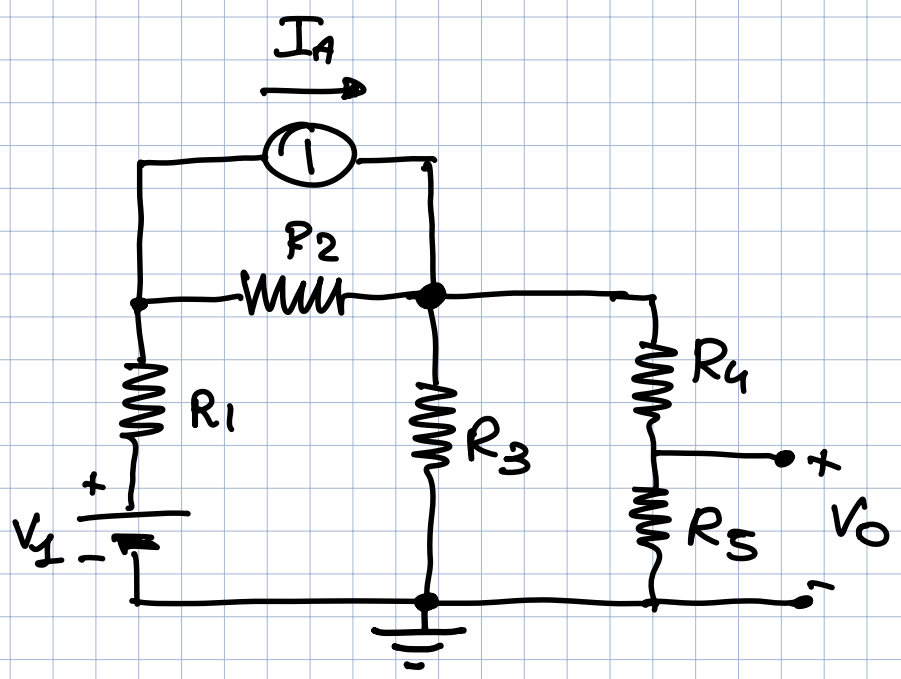
ESERCIZIO E1

DATI

$R_i = 2\,K\Omega$

$I_A = 1\,mA$

$V_1 = 12\,V$



ESERCIZIO E2

DATI

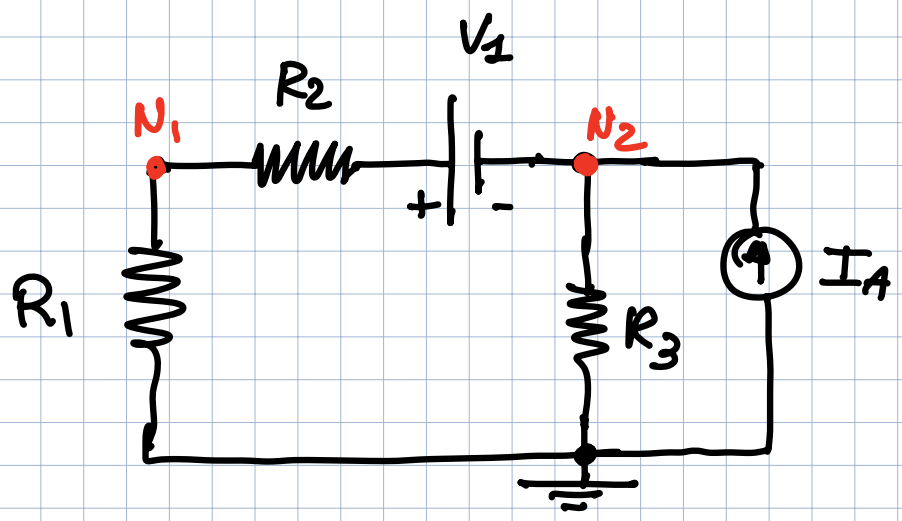
$R_1 = 8\,\Omega$

$R_2 = 2\,\Omega$

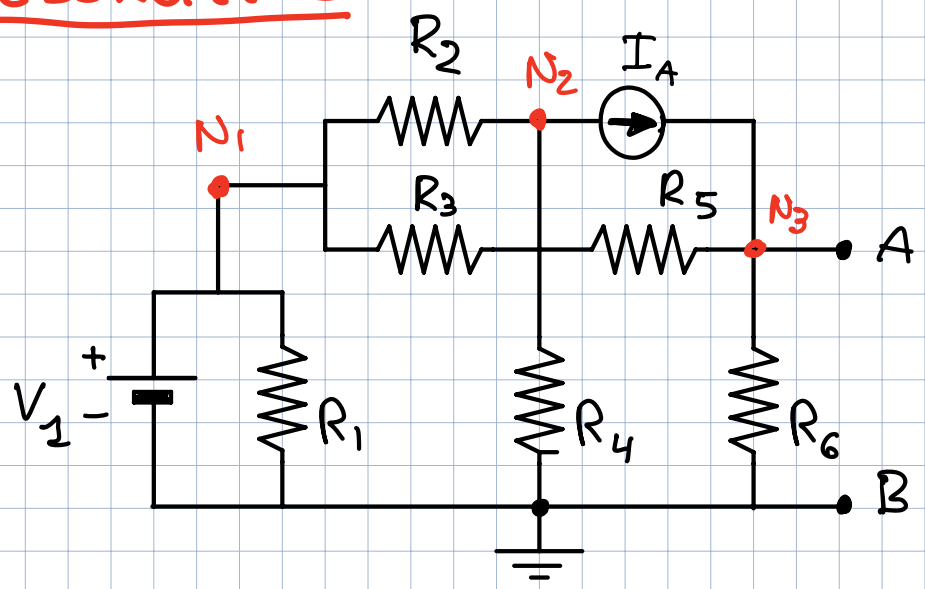
$R_3 = 4\,\Omega$

$I_A = 1\,A$

$V_1 = 12\,V$



ESERCIZIO 3



DATI

$R_1 = 10\,K\Omega$ $I_A = 10\,mA$

$R_2 = 4\,K\Omega$ $V_1 = 10\,V$

$R_3 = 4\,K\Omega$

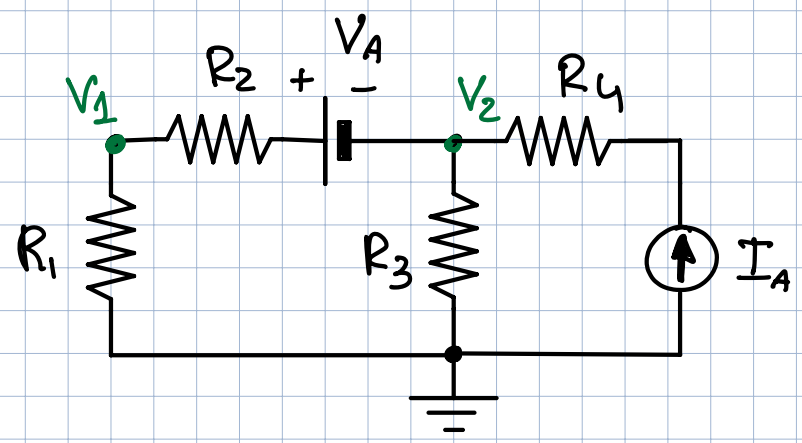
$R_4 = 2\,K\Omega$

$R_5 = 1\,K\Omega$

$R_6 = 2\,K\Omega$

PER QUESTO ESERCIZIO CALCOLARE INOLTRE IL GENERATORE EQUIVALENTE DI THEVENIN E NORTON VISTO DAI MORSETTI A-B

ESERCIZIO 4



$R_1 = 5\,K\Omega$ $V_A = 15\,V$

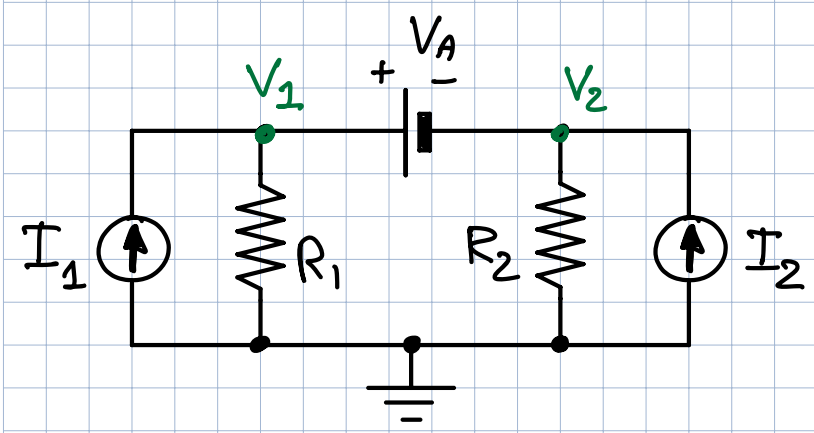
$R_2 = 10\,K\Omega$ $I_A = 5\,mA$

$R_3 = 5\,K\Omega$

$R_4 = 15\,K\Omega$

RISOLVERE IL CIRCUITO E CALCOLARE V_1 e V_2

ESERCIZIO 5



$R_1 = 10\Omega$ $V_A = 8V$

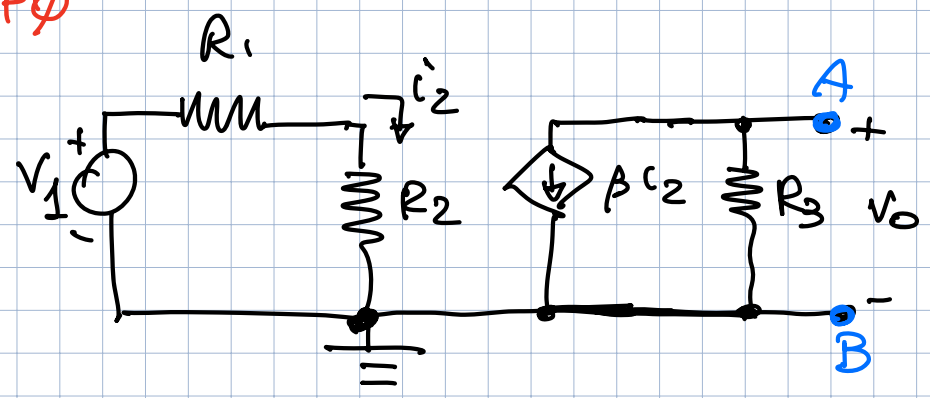
$R_2 = 5\Omega$ $I_1 = 5A$

$I_2 = 2A$

RISOLVERE IL CIRCUITO E CALCOLARE V_1 e V_2

RISOLVERE I SEGUENTI CIRCUITI CHE IMPIEGANO GENERATORI PILOTATI

EPØ



$V_1 = 10V$

$R_1 = 1 K\Omega$

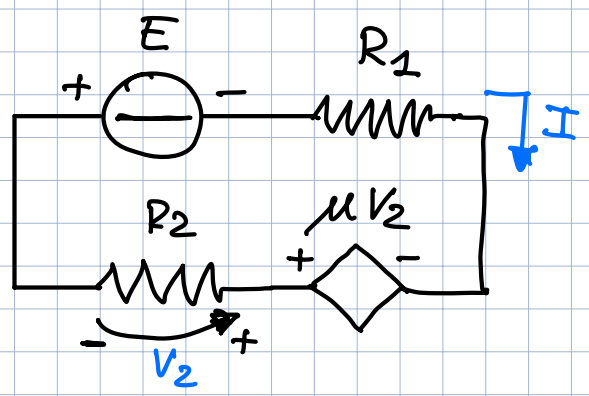
$R_2 = 4 K\Omega$

$R_3 = 1 K\Omega$

$\beta = 10$

TROVARE L'EQUIVALENTE DI THEVENIN E NORTON VISTO DA AB

EP1

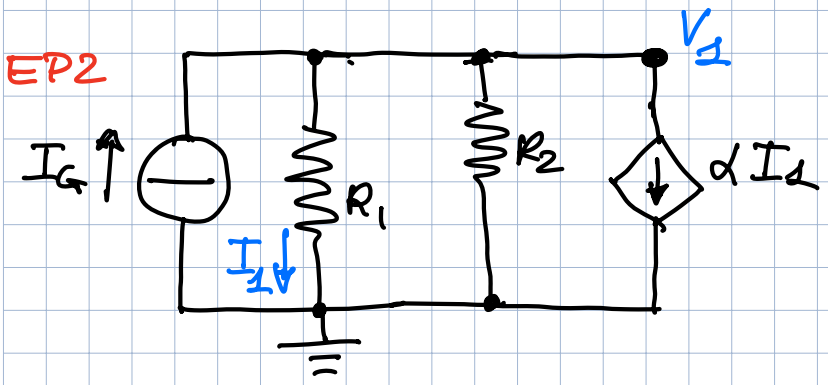


$R_1 = 4 K\Omega$ $E = 12V$

$R_2 = 2 K\Omega$ $\mu = 5$

CALCOLARE I, V_2

EP2

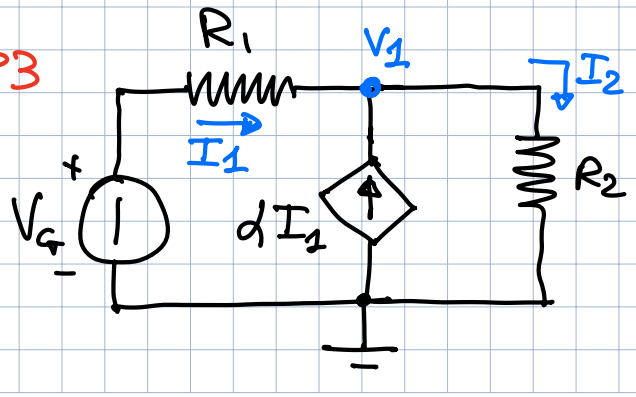


$R_1 = 4 K\Omega$ $I_G = 5 mA$

$R_2 = 2 K\Omega$ $\alpha = 2$

CALCOLARE I_1 e V_1

EP3

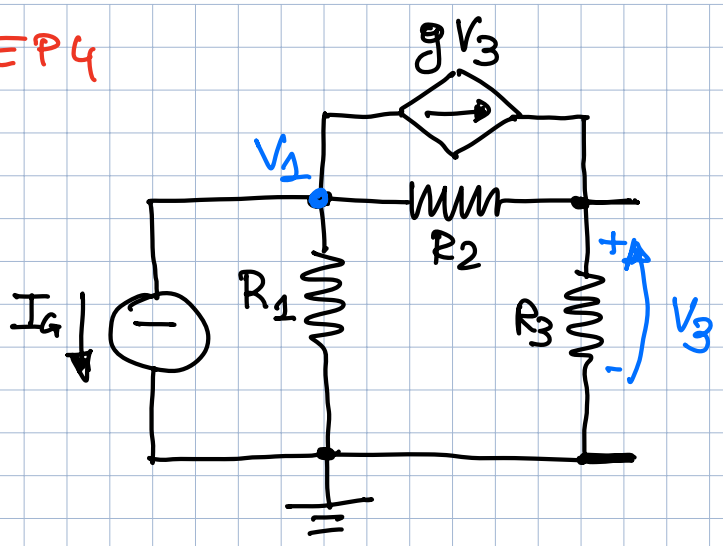


$R_1 = 4 K\Omega$ $V_G = 12V$

$R_2 = 2 K\Omega$ $\alpha = 3$

CALCOLARE I_1, I_2 e V_1

EP4



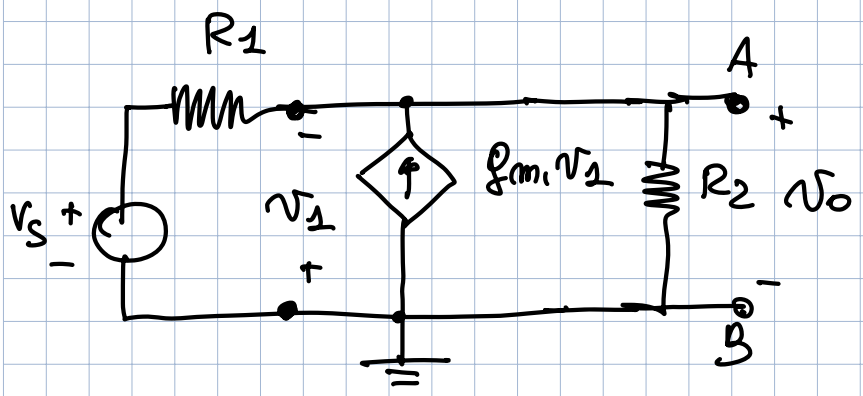
$R_1 = 2\,K\Omega$ $\quad I_G = 4\,mA$

$R_2 = 4\,K\Omega$ $\quad g = 2\,mS$

$R_3 = 2\,K\Omega$

CALCOLARE V_1, V_3 e LE CORRENTI SULLE 3 RESISTENZE.

ESERCIZIO EP5



$R_1 = 1\,K\Omega$ $\quad g_{m1} = 1\,mS$

$R_2 = 1\,K\Omega$ $\quad V_S = 1\,V$

TROVARE L'EQUIVALENTE DI THEVENIN E NORTON VISTO DA AB